MAINS IN, FUSE, SURGE; OUTLET (LIGHT always on, MOTOR controlled)

J1
IEC inlet
F1
T2A 250V
RV1
150VAC
L (fused)
N
PE
J2
duplex outlet: PE, LIGHT, N, MOTOR
MOTOR

TRIAC STAGE (after AN1662: triac in series with the motor)

MT1 (via K1)
R6
100
1W DNP
C2
47n
X2 DNP
Q1
BTA08-600BW
R5
330
GATE
R4
360 0.5W
U3
MOC3023
MT
NC
MT
A
NC
K
TRIAC FIRING
GND
R11
270

MOTOR-ENABLE RELAY

K1
G5LE-1A 5V
NO
COM
COIL+
COIL-
+5V
D1
1N4148
R12
100
Q2
2N7002
R13
100k
GND
RELAY DRIVER

ZERO-CROSS

R7
22k 0.5W
R8
22k 0.5W
U2
H11AA1
AC
NC
AC
C
B
E
+3V3
GND
R9
10k
ZERO-CROSS INPUT
C3
1n
GND

ISOLATED 5 V

F2
T250mA
PS1
IRM-05-5
L
N
+Vo
-Vo
+5V
GND
5 V BULK
C12
10u
C13
100n

ISOLATION BARRIER: U3, K1, U2 and PS1 straddle it; keep at least 6 mm creepage across it

LINE SIDE: LIVE, 120 V

CONTROL SIDE: isolated, SELV (3.3 V / 5 V)

OPERATOR INPUTS (RC-filtered ADC)

RV2
10k speed limit
R20
1k
LIMIT
C9
100n
J6
pressure pedal (FSR)
R19
1k
R18
10k
C8
100n
FSR
J5
travel pedal
R17
1k
C7
100n
PEDAL
J4
knee lever
R16
1k
C6
100n
KNEE

RP2350B CORE BOARD (pins by GPIO number)

A1 RP2350B core board
VBUS
GND
RUN
VSYS
GP47/ADC7
GP46/ADC6
GP45/ADC5
GP44/ADC4
GP43/ADC3
GP42/ADC2
GP41/ADC1
GP40/ADC0
ADC_VREF
3V3(OUT)
3V3_EN
GP0/UART0_TX
GP1/UART0_RX
GP2
GP3
GP4
GP5
GP6
GP7
GP8
GP9
GP10
GP11
GP12
GP13
D2
SS14
(Schottky)
FIRE
RELAY
ZC
HALL
STITCH
NEEDLE

SWITCHES (internal pull-ups)

SW1
needle up/down
C10
100n
SW2
single stitch
C11
100n

HAND-WHEEL HALL SENSOR (3-wire cable; one magnet, one pulse per stitch)

R14
10k
R15
100
C4
1n
U4
DRV5023 at hand wheel
VCC
OUT
GND
C5
100n
at sensor

NOTES. A concept drawn by Claude at Bob Drzyzgula's direction, October 2026: not built, not tested, values not checked.

1. After Freescale AN1662: zero-cross interrupt sets a timer alarm (or a PIO program); at the firing delay a ~100 us gate pulse fires the triac; hand-wheel period from edge timestamps; speed loop every half-cycle (120 Hz); stall watchdog. Added: isolation, relay, inputs.
2. Pins (RP2350B): FIRE GP4; RELAY GP5; ZC GP6; HALL GP7; STITCH GP10; NEEDLE GP12; knee, pedal, FSR, limit on ADC0-3 (GP40-43). ADC4-7 spare; GP0/GP1 UART0 for a debug line (or USB serial). Any GPIO will do for the digital signals; wire by GPIO number.
3. PS1 feeds VSYS through D2, so USB may stay plugged in (the control side is SELV). Flash by UF2 (BOOTSEL) or SWD. Drive GP4 at 8-12 mA.
4. Fire only after a zero-cross has been seen and the command is zero; close K1 last and open it first. Never scope the line side.
5. R7/R8: ~3.9 mA peak in the H11AA1; R11: ~8 mA into U3 from 3.3 V; R6/C2 fitted only if a snubberless triac false-fires.

Drafted by Claude at the direction of Robert Drzyzgula; not built or tested
After Freescale AN1662, with an isolated control side; RP2350B core board

Sheet: /
File: mcu_controller.kicad_sch

Title: VSM MCU phase controller (concept)

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